MASTERFUNCTIONALSKILLS

★ OFQUAL REGULATED STANDARDS

FUNCTIONAL SKILLS MATHEMATICS — Level 1 Essential

Maths Level 1 Revision Cards

Visual Rules • Exam Challenges • Step-by-Step Solutions

- ★ Perimeter, Area & Metric Conversions
- ★ Fractions, Decimals, Percentages & BODMAS
- ★ Averages (Mean & Range) & Probability
- ★ Targeted Exam Hacks & Common Traps

A6 Postcard Cardstock (148 × 105 mm)

Turn over for Deck Contents & Checklist ➔

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1. Place Value up to 1,000,000

NUMBER SYSTEM

Millions, Thousands & Ones

Numbers are grouped in 3-digit periods separated by commas or spaces: Millions (M), Hundred Thousands (HTh), Ten Thousands (TTh), Thousands (Th), Hundreds (H), Tens (T), Units (U).

482,519 (Four Hundred & Eighty-Two Thousand...)

4 HTh	8 TTh	2 Th	5 H	1 T	9 U

The digit '8' has a value of 80,000 (Eighty Thousand)

★ Golden Rule: Read large numbers in groups of three digits from left to right!

⚠ Watch Out: Don't confuse 'Eight Thousand' (8,000) with 'Eighty Thousand' (80,000).

DfE L1.N1 • Level 1 — Turn over for Practice Challenge ➔ — A6 Pocket Card

2. Four Operations with Whole Numbers

NUMBER SYSTEM

Column Methods & Checking

Addition & Subtraction: Line up unit digits strictly on the right. Carry over when sum ≥ 10; borrow from next column to the left for subtraction. Multiplication: Long column. Division: Short 'bus stop'.

SUBTRACTION

5 2 4
- 1 6 8
= 3 5 6

SHORT DIVISION

6) 8 5 ²4
= 1 4 2

★ Golden Rule: Always check subtraction by adding your answer to the smaller number!

⚠ Watch Out: Don't subtract smaller from larger in the same column! If top digit is smaller, borrow 1 from left.

DfE L1.N2 • Level 1 — Turn over for Practice Challenge ➔ — A6 Pocket Card

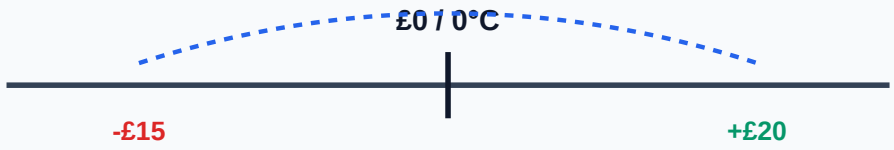
3. Negative Numbers (Temp & Debt)

NUMBER SYSTEM

Below Zero & Bank Overdrafts

Numbers below 0 have a minus sign (-). Moving RIGHT on the number line = getting warmer / paying in money (+). Moving LEFT = getting colder / spending money (-).

Difference = £35



-£15 is DEEPER in debt than -£5

★ Golden Rule: -8°C is COLDER than -3°C!

⚠ Watch Out: Overdraft balance: If balance is -£25 and you spend £10, new balance is -£35 (NOT -£15).

DfE L1.N3 • Level 1 — Turn over for Practice Challenge ➔ — A6 Pocket Card

Challenge: Place Value up to 1,000,000

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE L1.N1

What is the value of the digit 7 in the number 574,820?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

The digit 7 is in the ten thousands column, so its value is 70,000 (Seventy Thousand).

📋 DECK CONTENTS & REVISION CHECKLIST

38 Revision Cards

Tick [✓] Read when you understand the rule, and [★] Solved when you answer the challenge! Pearson & C&G Aligned

1. Place Value up to 1,000,000 (DfE L1.N1)
2. Four Operations with Whole Numbers (DfE L1.N2)
3. Negative Numbers (Temp & Debt) (DfE L1.N3)
4. Rounding to 10, 100, 1000 (DfE L1.N4)
5. Decimals: Place Value & Ordering (DfE L1.N5)
6. Operations with Decimals & Money (DfE L1.N6)
7. FDP Common Benchmarks (DfE L1.N7)
8. Simplifying Fractions (DfE L1.N8)
9. Fraction of an Amount (Basic) (DfE L1.N9)
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11. Simple Ratios (2 Parts) (DfE L1.N11)
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15. Perimeter of Rectangle & Square (DfE L1.MSS15)
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17. Area of a Rectangle (DfE L1.MSS16)
18. Area of a Triangle (DfE L1.MSS17)
19. Volume of a Cuboid (DfE L1.MSS18)
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21. Metric Weight Conversions (DfE L1.MSS20)
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28. Scale Drawings on Grid Paper (DfE L1.MSS27)
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30. Median (Middle Value) (DfE L1.HD29)
31. Mode (Most Frequent) (DfE L1.HD30)
32. Range (Spread of Data) (DfE L1.HD31)
33. Tally Charts & Frequency (DfE L1.HD32)
34. Bar Charts & Scales (DfE L1.HD33)
35. Pictograms (Using a Key) (DfE L1.HD34)
36. Line Graphs (Trends over Time) (DfE L1.HD35)
37. Probability Words & Scale (DfE L1.HD36)
38. Comparing Two Data Sets (L1) (DfE L1.HD37)

💡 **Tutor Golden Tip (Abdul):** Do not flip the card immediately! Attempt the Quick Challenge on the back using a dry-wipe pen or scratchpad first. Verify with the step-by-step model steps only after completing your working.

Challenge: Negative Numbers (Temp & Debt)

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE L1.N3

A bank account is overdrawn by -£45. A deposit of £70 is paid in. What is the new bank balance?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

New balance = -45 + 70 = +£25.

Challenge: Four Operations with Whole Numbers

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE L1.N2

Calculate: 735 - 289. Show your check using addition.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

735 - 289 = 446. Check: 446 + 289 = 735 ✓.

4. Rounding to 10, 100, 1000

NUMBER SYSTEM

Look at the Neighbour on the Right

Find the rounding place: Look at the next digit to the right: If 5 or more (5, 6, 7, 8, 9), ROUND UP. If 4 or less (0, 1, 2, 3, 4), ROUND DOWN (stay same). Fill trailing places with 0.

Round 4,763 to the nearest 100:

Hundreds digit is 7. Neighbour is 6.

6 is ≥ 5 → Round up to 4,800

⋆ **Golden Rule: 5 to 9 climbs the vine; 0 to 4 stays on the floor!**

⚠️ **Watch Out: Don't delete digits! When rounding 3,428 to the nearest 10, it becomes 3,430, NOT 343.**

5. Decimals: Place Value & Ordering

NUMBER SYSTEM

Tenths, Hundredths & Thousandths

To compare decimals: Line up decimal points vertically and pad with trailing zeros so all numbers have the same number of digits after the dot (e.g. 0.4 becomes 0.400).

Which is bigger: 0.35 or 0.4?

Pad with a zero: 0.4 → 0.40

Compare 35 hundredths vs 40 hundredths

0.4 is BIGGER than 0.35!

⋆ **Golden Rule: Add trailing zeroes to level the playing field before comparing decimals!**

⚠️ **Watch Out: 0.35 has more digits than 0.4, but 0.4 is LARGER! (0.40 > 0.35).**

6. Operations with Decimals & Money

NUMBER SYSTEM

Line Up the Decimal Points

Always align the decimal points directly under each other when adding or subtracting money. Money must ALWAYS have TWO decimal places (e.g. £4.50, NEVER £4.5).

LINE UP THE DOTS

£14.50

+£3.85

£18.35

⋆ **Golden Rule: Always write pounds with 2 decimal places: £6.20, NOT £6.2!**

⚠️ **Watch Out: Don't mix pounds and pence without converting! £5 + 40p = £5.40, NOT £45.**

7. FDP Common Benchmarks

NUMBER SYSTEM

Halves, Quarters & Tenths

1/2 = 0.5 = 50%. 1/4 = 0.25 = 25%. 3/4 = 0.75 = 75%. 1/10 = 0.1 = 10%. 1/5 = 0.2 = 20%. Memorising these visual equivalents saves time in exams.

Fraction	Decimal	Percentage
1/2	0.5	50%
1/4	0.25	25%
3/4	0.75	75%

⋆ **Golden Rule: 1/5 is 20% (0.2), NOT 15%!**

⚠️ **Watch Out: Don't confuse 0.1 (10%) with 0.01 (1%).**

Challenge: Decimals: Place Value & Ordering

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE L1.N5

Put in order from smallest to largest: 0.08, 0.8, 0.75, 0.079.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Pad to 3 decimal places: 0.080, 0.800, 0.750, 0.079. Order: 0.079, 0.08, 0.75, 0.8.

Challenge: Rounding to 10, 100, 1000

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE L1.N4

Round 18,450: (a) To the nearest 1,000. (b) To the nearest 100.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

(a) To nearest 1,000: 4 is < 5 → 18,000. (b) To nearest 100: 5 is ≥ 5 → 18,500.

Challenge: FDP Common Benchmarks

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE L1.N7

Which is greater: 0.3 or 1/4? Show your working.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

1/4 = 0.25. Since 0.30 > 0.25, 0.3 is greater.

Challenge: Operations with Decimals & Money

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE L1.N6

You buy items costing £3.45, £8.20, and £0.95. You pay with a £20 note. How much change do you receive?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Total = £3.45 + £8.20 + £0.95 = £12.60. Change = £20.00 - £12.60 = £7.40.

8. Simplifying Fractions

NUMBER SYSTEM

Divide Top and Bottom Equally

To simplify a fraction, divide both the numerator (top) and denominator (bottom) by their Highest Common Factor (HCF). Keep dividing until no common factors remain.

Simplify 12/18

Divide top and bottom by 6 (HCF):

12 ÷ 6 / 18 ÷ 6 = 2/3

⋆ Golden Rule: Whatever you do to the top, you MUST do to the bottom!

⚠ Watch Out: Never subtract numbers to simplify fractions — you can ONLY divide or multiply.

9. Fraction of an Amount (Basic)

NUMBER SYSTEM

Divide by Denominator, Times by Numerator

To find a unit fraction (1/n): Divide by n. To find a non-unit fraction (e.g. 3/4): 1. Divide by 4 to get 1 quarter. 2. Multiply result by 3 to get 3 quarters.

Find 3/4 of 48 kg

Step 1: 48 ÷ 4 = 12 kg (1/4)

Step 2: 12 × 3 = 36 kg

⋆ Golden Rule: Divide by bottom, multiply by top!

⚠ Watch Out: Don't stop after step 1! Dividing by 4 gives only ONE quarter (12 kg), you need THREE.

10. Percentages (Building Blocks)

NUMBER SYSTEM

10%, 5%, 1% Breakdown Method

Non-calculator percentage hacks: 10% = Divide by 10. 5% = Half of 10%. 1% = Divide by 100. 50% = Half (÷ 2). Combine blocks to find any percentage (e.g. 15% = 10% + 5%).

Find 15% of £80 without calculator

10% of £80 = £80 ÷ 10 = £8.00

5% of £80 = Half of £8 = £4.00

15% = £8.00 + £4.00 = £12.00

⋆ Golden Rule: Find 10% first — it unlocks 5%, 20%, 30%, and 15% instantly!

⚠ Watch Out: Don't divide by 15 to find 15%! You find 10% and add 5%.

11. Simple Ratios (2 Parts)

NUMBER SYSTEM

Add Parts, Divide Total, Multiply

To share £60 in ratio 1 : 3: 1. Add parts: 1 + 3 = 4 parts. 2. Divide total: £60 ÷ 4 = £15 (value of 1 part). 3. Multiply: Share 1 = £15, Share 2 = 3 × £15 = £45.

Share £60 in ratio 1 : 3

£15	£15	£15	£15

Result = £15 and £45 (Total = £60 ✓)

⋆ Golden Rule: Ratios must always be simplified to lowest whole numbers!

⚠ Watch Out: Keep the order of the ratio correct: Person A : Person B must match the order in the text.

Challenge: Fraction of an Amount (Basic)

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE L1.N9

A community group has 60 members. 2/5 of them are volunteers. How many volunteers are there?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

60 ÷ 5 = 12 (1/5). 12 × 2 = 24 volunteers.

Challenge: Simplifying Fractions

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE L1.N8

Write 24/36 in its simplest form.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Divide by 12: 24 ÷ 12 = 2. 36 ÷ 12 = 3. Simplest form = 2/3.

Challenge: Simple Ratios (2 Parts)

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE L1.N11

Simplify the ratio 15 : 25 to its simplest form.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Divide both by 5: 15 ÷ 5 = 3, 25 ÷ 5 = 5. Simplified ratio = 3 : 5.

Challenge: Percentages (Building Blocks)

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE L1.N10

Work out 35% of £240 without using a calculator.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

10% = £24. 30% = 24 × 3 = £72. 5% = half of £24 = £12. 35% = £72 + £12 = £84.

12. Direct Proportion (Recipes)

NUMBER SYSTEM

Unitary Method: Find for 1 First

To scale a recipe: 1. Divide ingredients by the original number of servings to find the amount for 1 person. 2. Multiply by the new number of people needed.

Recipe for 4 people: 200g flour. Need for 6 people?

Step 1 (For 1 person): 200g ÷ 4 = 50g

Step 2 (For 6 people): 50g × 6 = 300g flour

★ Golden Rule: Find the quantity for ONE person first, then multiply!

⚠️ Watch Out: Don't just add 2 to the flour because 6 is 4 + 2 — you must scale proportionally!

13. Multiples, Factors & Primes

NUMBER SYSTEM

Definitions & Factor Pairs

Multiple = In the times table (Multiples of 5: 5, 10, 15...). Factor = Numbers that divide in exactly with no remainder. Prime = EXACTLY two factors: 1 and itself (2, 3, 5, 7, 11...).

FACTORS OF 12

1 × 12

2 × 6

3 × 4

PRIME NUMBERS < 20

2, 3, 5, 7, 11, 13, 17, 19

⚠️ 1 is NOT prime!

2 is the ONLY even prime

★ Golden Rule: 1 is NOT a prime number because it only has 1 factor!

⚠️ Watch Out: Factors are smaller (or equal); multiples are larger (or equal).

14. Inverse Operations for Checking

NUMBER SYSTEM

Reverse Your Calculation for 1 Mark

Pearson exams award a dedicated mark for 'Show a check of your answer': Addition check with SUBTRACTION. Subtraction check with ADDITION. Multiplication check with DIVISION. Division check with MULTIPLICATION.

PEARSON SECTION A CHECKING HACK

Main Calculation: 45 × 8 = 360

Written Check: 360 ÷ 8 = 45 ✓

★ Golden Rule: Use reverse operations (add ↔ subtract, multiply ↔ divide) to check!

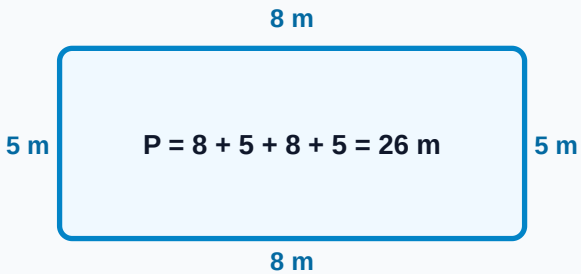
⚠️ Watch Out: Don't repeat the same sum — examiners require the inverse operation!

15. Perimeter of Rectangle & Square

MEASURES & SPACE

Add All Four Outside Edges

Perimeter is the total boundary distance around a 2D shape. Rectangle: Perimeter = Length + Width + Length + Width = 2l + 2w. Units are linear (cm, m).



★ Golden Rule: Perimeter is the border fence around the shape!

⚠️ Watch Out: Don't just add 8 + 5 = 13! A rectangle has FOUR sides, so you must add all 4 sides.

Challenge: Multiples, Factors & Primes

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE L1.N13

List all the factors of 20.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Factor pairs: 1 × 20, 2 × 10, 4 × 5. Factors of 20: 1, 2, 4, 5, 10, 20.

Challenge: Direct Proportion (Recipes)

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE L1.N12

A soup recipe for 8 people needs 600 ml of stock. How much stock is needed for 12 people?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

For 1 person: 600 ÷ 8 = 75 ml. For 12 people: 75 × 12 = 900 ml.

Challenge: Perimeter of Rectangle & Square

MEASURES & SPACE

🎯 QUICK EXAM CHALLENGE — DfE L1.MSS15

A square garden has a side length of 7.5 metres. How many metres of fencing are needed to go all the way around?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

A square has 4 equal sides. Perimeter = 4 × 7.5 = 30 metres.

Challenge: Inverse Operations for Checking

NUMBER SYSTEM

🎯 QUICK EXAM CHALLENGE — DfE L1.N14

You calculated: 156 + 89 = 245. Write down a valid check of your working using inverse operations.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

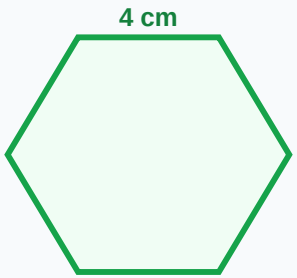
Check: 245 - 89 = 156 (or 245 - 156 = 89).

16. Perimeter of Regular Polygons

MEASURES & SPACE

Side Length × Number of Sides

A 'regular' polygon has sides that are ALL EQUAL in length: Equilateral Triangle = 3 × side. Pentagon = 5 × side. Hexagon = 6 × side. Octagon = 8 × side.



Regular Hexagon

6 equal sides of 4 cm

Perimeter = 6 × 4 = 24 cm

★ **Golden Rule: Regular means all sides and angles are identical!**

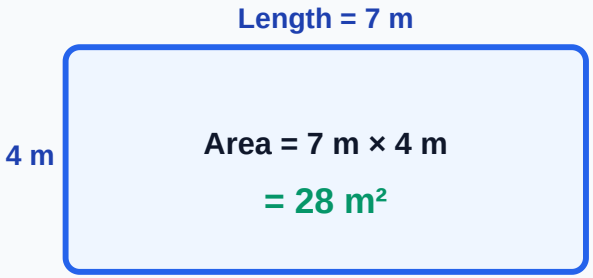
⚠️ **Watch Out: Don't count the sides incorrectly: Hexagon = 6, Octagon = 8, Pentagon = 5.**

17. Area of a Rectangle

MEASURES & SPACE

Area = Length × Width

Area measures the flat 2D space inside a shape. Multiply length by width. Answers are ALWAYS written in square units (cm², m², mm²).



★ **Golden Rule: Area is flat surface (carpet/grass), measured in square units (m²)!**

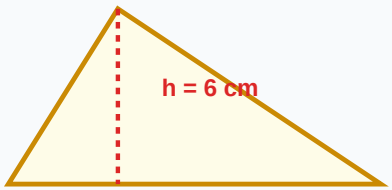
⚠️ **Watch Out: Don't confuse Area (L × W) with Perimeter (L + W + L + W).**

18. Area of a Triangle

MEASURES & SPACE

Area = (Base × Height) ÷ 2

1. Multiply base by perpendicular vertical height. 2. HALVE the answer (divide by 2) because a triangle is half of a rectangle.



Area = (10 × 6) ÷ 2 = 30 cm²

★ **Golden Rule: Always remember to divide by 2 for a triangle!**

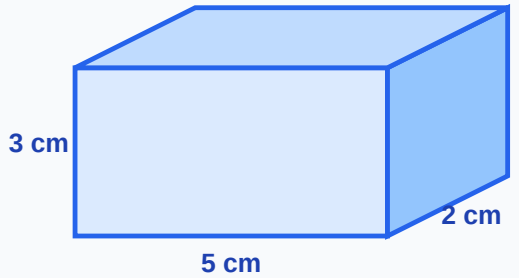
⚠️ **Watch Out: Never use the slanted side length for area — only use vertical perpendicular height.**

19. Volume of a Cuboid

MEASURES & SPACE

Volume = Length × Width × Height

Volume measures the 3D space inside a solid box. Multiply all three dimensions together: Length × Width × Height. Written in cubic units (cm³, m³).



V = 5 × 2 × 3

= 30 cm³

★ **Golden Rule: Volume is 3D capacity, measured in cubic units (cm³ or m³)!**

⚠️ **Watch Out: Don't add the three dimensions — you must MULTIPLY them.**

Challenge: Area of a Rectangle

MEASURES & SPACE

🎯 QUICK EXAM CHALLENGE — DfE L1.MSS16

A living room floor is 6.5 m long and 4 m wide. How many square metres of carpet are needed to cover it?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Area = 6.5 × 4 = 26 m².

Challenge: Perimeter of Regular Polygons

MEASURES & SPACE

🎯 QUICK EXAM CHALLENGE — DfE L1.MSS15

A stop sign is a regular octagon with each side measuring 25 cm. What is the total perimeter in metres?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Octagon has 8 sides. Perimeter = 8 × 25 cm = 200 cm = 2.0 metres.

Challenge: Volume of a Cuboid

MEASURES & SPACE

🎯 QUICK EXAM CHALLENGE — DfE L1.MSS18

A packing box is 40 cm long, 25 cm wide, and 20 cm high. What is its volume in cm³?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Volume = 40 × 25 × 20 = 1,000 × 20 = 20,000 cm³.

Challenge: Area of a Triangle

MEASURES & SPACE

🎯 QUICK EXAM CHALLENGE — DfE L1.MSS17

A triangular banner has a base of 12 cm and a vertical height of 8 cm. What is its area?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Area = (12 × 8) ÷ 2 = 96 ÷ 2 = 48 cm².

20. Metric Length Conversions

MEASURES & SPACE

mm • cm • m • km

1 cm = 10 mm. 1 m = 100 cm. 1 km = 1,000 m. Big to small unit = MULTIPLY (×). Small to big unit = DIVIDE (÷).

METRIC LENGTH STEPS

1 km = 1,000 m (e.g. 3.2 km = 3,200 m)

1 m = 100 cm (e.g. 4.5 m = 450 cm)

1 cm = 10 mm (e.g. 8 cm = 80 mm)

★ Golden Rule: Metres to cm: MULTIPLY by 100. cm to metres: DIVIDE by 100.

⚠️ Watch Out: Don't confuse mm and cm: 45 mm is 4.5 cm, NOT 450 cm.

21. Metric Weight Conversions

MEASURES & SPACE

1 kg = 1,000 g | 1 tonne = 1,000 kg

To convert kg to g: MULTIPLY by 1,000. To convert g to kg: DIVIDE by 1,000. For medicine: 1 g = 1,000 mg.

MASS: KILOGRAMS & GRAMS

kg → (× 1,000) → grams

grams → (÷ 1,000) → kg

Example: 2.5 kg = 2,500 g

★ Golden Rule: 'Kilo' always means 1,000!

⚠️ Watch Out: Half a kilogram is 500 grams, NOT 50 grams.

22. Metric Capacity Conversions

MEASURES & SPACE

1 Litre = 1,000 ml

Litres to millilitres: MULTIPLY by 1,000. Millilitres to litres: DIVIDE by 1,000. Common amounts: 1/2 Litre = 500 ml. 1/4 Litre = 250 ml. 3/4 Litre = 750 ml.

CAPACITY: LITRES & MILLILITRES

1 L = 1,000 ml

1/4 L = 250 ml • 1/2 L = 500 ml • 3/4 L = 750 ml

e.g. 1.8 Litres = 1,800 ml

★ Golden Rule: Milli means one-thousandth!

⚠️ Watch Out: Don't divide by 100 — there are 1,000 millilitres in a litre.

23. Time: 12-Hour & 24-Hour Clock

MEASURES & SPACE

Converting & Timetables

To convert PM time to 24-hour: ADD 12 hours (e.g. 4:15 pm + 12 = 16:15). To convert 24-hour to PM: SUBTRACT 12 hours (e.g. 21:40 - 12 = 9:40 pm).

12-HOUR CLOCK	24-HOUR CLOCK
8:30 pm	20:30
Add 12 hours for PM	Always 4 digits, NO 'pm'

★ Golden Rule: Never write 'pm' after a 24-hour time (e.g. 20:30, NOT 20:30 pm)!

⚠️ Watch Out: 12:30 pm is afternoon (lunchtime) = 12:30. 12:30 am is midnight = 00:30.

Challenge: Metric Weight Conversions

MEASURES & SPACE

🎯 QUICK EXAM CHALLENGE — DfE L1.MSS20

A postal parcel weighs 1,450 grams. What is its weight in kilograms?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

1,450 ÷ 1,000 = 1.45 kg.

Challenge: Metric Length Conversions

MEASURES & SPACE

🎯 QUICK EXAM CHALLENGE — DfE L1.MSS19

Convert 275 cm into metres.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

275 ÷ 100 = 2.75 metres.

Challenge: Time: 12-Hour & 24-Hour Clock

MEASURES & SPACE

🎯 QUICK EXAM CHALLENGE — DfE L1.MSS22

A train departs at 14:45 and arrives at 16:10. How long was the journey in hours and minutes?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

From 14:45 to 15:00 is 15 mins. From 15:00 to 16:10 is 1 hr 10 mins. Total = 1 hour 25 minutes.

Challenge: Metric Capacity Conversions

MEASURES & SPACE

🎯 QUICK EXAM CHALLENGE — DfE L1.MSS21

A bottle contains 2 Litres of water. 350 ml is poured out. How many ml remain in the bottle?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

2 Litres = 2,000 ml. Remaining = 2,000 - 350 = 1,650 ml (or 1.65 L).

24. Money: Change & Best Buys

MEASURES & SPACE

Multi-Item Totals & Change

1. Multiply quantities by unit price. 2. Sum all items carefully. 3. Change = Amount paid - Total bill. Compare deals by finding price for the same amount.

Shopping Bill & Change

3 sandwiches @ £2.50 = £7.50

2 drinks @ £1.25 = £2.50 → Total = £10.00

Paid with £20 note → Change = £20 - £10 = £10.00

⋆ Golden Rule: Always write both decimal places for currency: £10.00, NOT £10!

⚠ Watch Out: Don't add items without multiplying by quantity first.

25. Properties of 2D Shapes

MEASURES & SPACE

Sides, Angles & Symmetry

Right Angle = 90°. Triangle angles sum to 180°. Quadrilateral angles sum to 360°. Parallel lines never meet. Perpendicular lines meet at 90°.

TRIANGLES (180°)

Equilateral: 3 equal sides

Isosceles: 2 equal sides

Scalene: 0 equal sides

QUADRILATERALS (360°)

Square: 4 equal sides, 90°

Rectangle: Opposite equal

Trapezium: 1 pair parallel

⋆ Golden Rule: All angles in ANY triangle add up to 180°!

⚠ Watch Out: A rhombus has 4 equal sides, but its angles are NOT 90° (a square has 90° angles).

26. Properties of 3D Solids

MEASURES & SPACE

Faces, Edges & Vertices

Face = Flat or curved surface. Edge = Line where two faces meet. Vertex (Vertices) = Corner point where edges meet. A cube/cuboid has 6 faces, 12 edges, 8 vertices.

CUBOID 3D PROPERTIES

Faces = 6 flat rectangular faces

Edges = 12 straight lines

Vertices = 8 corner points

⋆ Golden Rule: Vertices = Corners • Edges = Creases • Faces = Surfaces.

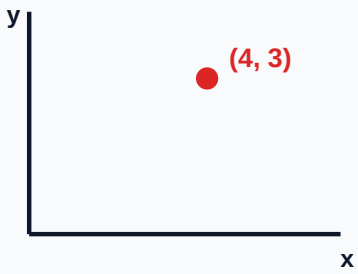
⚠ Watch Out: A cylinder has 3 faces (2 flat circles, 1 curved), 2 edges, and 0 vertices!

27. Coordinates (x, y)

MEASURES & SPACE

Along the Corridor, Up the Stairs

Coordinates are written as (x, y): 1. First number (x) goes HORIZONTALLY along the bottom axis. 2. Second number (y) goes VERTICALLY up the side axis. Origin is (0, 0).



Point (4, 3):

4 along (x), 3 up (y)

⋆ Golden Rule: 'x' comes before 'y' in the alphabet, and in coordinates (x, y)!

⚠ Watch Out: Don't mix up (2, 5) with (5, 2) — always read the bottom horizontal axis first.

Challenge: Properties of 2D Shapes

MEASURES & SPACE

🎯 QUICK EXAM CHALLENGE — DfE L1.MSS24

Two angles in a triangle are 65° and 45°. What is the size of the third angle?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Sum of known angles = 65° + 45° = 110°. Third angle = 180° - 110° = 70°.

Challenge: Money: Change & Best Buys

MEASURES & SPACE

🎯 QUICK EXAM CHALLENGE — DfE L1.MSS23

You buy 4 notebooks at £1.75 each and a pack of pens for £2.80. You pay with a £10 note. What change do you get?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Notebooks = 4 × £1.75 = £7.00. Total = £7.00 + £2.80 = £9.80. Change = £10.00 - £9.80 = £0.20 (20p).

Challenge: Coordinates (x, y)

MEASURES & SPACE

🎯 QUICK EXAM CHALLENGE — DfE L1.MSS26

What are the coordinates of the origin where the two axes cross?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

The coordinates of the origin are (0, 0).

Challenge: Properties of 3D Solids

MEASURES & SPACE

🎯 QUICK EXAM CHALLENGE — DfE L1.MSS25

How many vertices (corners) does a triangular prism have?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

A triangular prism has 6 vertices (3 at each triangular end).

28. Scale Drawings on Grid Paper

MEASURES & SPACE

Count Squares & Use the Key

Always check the scale key (e.g. 1 square = 1 metre, or 1 square = 2 metres). Real dimension = Number of squares counted × Scale key value.



Key: 1 sq = 2 metres

Room is 3 sq × 2 sq

Actual: 6 m by 4 m

⋆ **Golden Rule: Always read the scale key: 1 square is rarely 1 cm in real life!**

⚠️ **Watch Out: If 1 square = 2 metres, 4 squares = 8 metres, NOT 4 metres.**

29. Mean Average (Step-by-Step)

HANDLING DATA

Add All Up, Divide by Count

1. Add all numbers together to find total sum. 2. Count how many numbers there are. 3. Divide total sum by the count. Mean is the fair share.

Find Mean of: 6, 8, 12, 14

Step 1: Total = 6 + 8 + 12 + 14 = 40

Step 2: There are 4 numbers

Step 3: 40 ÷ 4 = 10

⋆ **Golden Rule: Divide by how many items you added up!**

⚠️ **Watch Out: Don't forget to press equals (=) before dividing on your calculator, or use brackets.**

30. Median (Middle Value)

HANDLING DATA

Line Up Low to High, Pick Middle

1. Put numbers in numerical order from smallest to largest. 2. Cross off one from each end until you reach the middle number. If two remain in middle, find their midpoint.

Raw Data: 9, 3, 7, 2, 8

1. Put in order: 2, 3, 7, 8, 9

Median = 7

⋆ **Golden Rule: You MUST sort the numbers from smallest to largest first!**

⚠️ **Watch Out: Never pick the middle number of the unsorted list (in 9, 3, 7, 2, 8, '7' happens to be middle, but never rely on luck!).**

31. Mode (Most Frequent)

HANDLING DATA

The Most Common Item

Mode is the number or category that appears MOST OFTEN in the data. There can be one mode, more than one mode (bimodal), or no mode if all values appear once.

Shoe Sizes: 5, 6, 6, 7, 6, 8, 5

Size 5 appears 2 times • Size 6 appears 3 times

Mode = Size 6 ⋆

⋆ **Golden Rule: 'MO'de = 'MO'st often!**

⚠️ **Watch Out: The mode can be words (e.g. Blue, Coffee) — it is the only average that works for non-numbers.**

Challenge: Mean Average (Step-by-Step)

HANDLING DATA

🎯 QUICK EXAM CHALLENGE — DfE L1.HD28

Five friends score 14, 18, 12, 20, and 16 in a quiz. What is their mean score?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Total = 14 + 18 + 12 + 20 + 16 = 80. Count = 5. Mean = 80 ÷ 5 = 16.

Challenge: Scale Drawings on Grid Paper

MEASURES & SPACE

🎯 QUICK EXAM CHALLENGE — DfE L1.MSS27

On a grid plan where 1 square = 0.5 metres, a desk is drawn 4 squares long. What is the real length of the desk?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Real length = 4 × 0.5 m = 2.0 metres.

Challenge: Mode (Most Frequent)

HANDLING DATA

🎯 QUICK EXAM CHALLENGE — DfE L1.HD30

What is the modal value of this dataset: 4, 7, 2, 7, 5, 9, 7, 2?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

7 appears three times (more than any other number). Mode = 7.

Challenge: Median (Middle Value)

HANDLING DATA

🎯 QUICK EXAM CHALLENGE — DfE L1.HD29

Find the median of: 15, 6, 21, 9, 12.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Order: 6, 9, 12, 15, 21. Middle value is 12 (Median = 12).

32. Range (Spread of Data)

HANDLING DATA

Range = Highest - Lowest

Range is NOT an average — it measures how spread out the numbers are. Subtract the smallest number from the largest number. Smaller range = more consistent.

Temperatures: 12°C, 25°C, 18°C, 14°C

Highest = 25°C • Lowest = 12°C

Range = 25 - 12 = 13°C

⋆ Golden Rule: Range = Highest minus Lowest!

⚠ Watch Out: Don't write '12 to 25' — the range is a SINGLE number obtained by subtracting (13).

33. Tally Charts & Frequency

HANDLING DATA

Bundles of 5 (4 Vertical + 1 Diagonal Gate)

Tallies are grouped in bundles of five: four vertical marks (||||) with the fifth mark crossing diagonally as a gate across the bundle. Frequency is the total count written as a number.

TALLY SYSTEM (5-BAR GATE)

Item: Car  Freq: 8

Count tallies in 5s: 5 + 3 = 8

⋆ Golden Rule: The 5th tally stroke always gates diagonally across the bundle of 4!

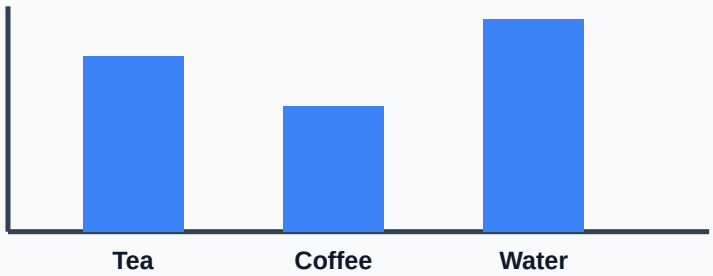
⚠ Watch Out: Check your raw data twice by crossing off each item as you record its tally mark.

34. Bar Charts & Scales

HANDLING DATA

Equal Widths, Equal Gaps, Labelled Axes

To draw a bar chart for full marks: 1. Evenly spaced linear scale on vertical axis starting at 0. 2. Label both axes (e.g. 'Days' and 'Frequency'). 3. Equal width bars with equal gaps between them.



⋆ Golden Rule: Bars must NOT touch each other in a categorical bar chart!

⚠ Watch Out: Check what 1 grid square represents on the vertical axis (e.g. 1 square might be 2 or 5 units).

35. Pictograms (Using a Key)

HANDLING DATA

Always Check the Symbol Key

A pictogram uses pictures or symbols to represent quantities. Look at the key: 1 whole symbol = n items. A half symbol = n ÷ 2. A quarter symbol = n ÷ 4.

KEY: [⋆] = 4 BOOKS

Monday: [⋆] [⋆] = 2 × 4 = 8 books

Tuesday: [⋆] [⋆] [half] = 4 + 4 + 2 = 10 books

⋆ Golden Rule: Always check the KEY first — one picture is rarely just 1 item!

⚠ Watch Out: Don't count the number of symbols without multiplying by the key value.

Challenge: Tally Charts & Frequency

HANDLING DATA

🎯 QUICK EXAM CHALLENGE — DfE L1.HD32

A tally chart shows three 5-bar tally gates and 2 single vertical strokes. What frequency number does this represent?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Three bundles of 5 plus 2: (3 × 5) + 2 = 15 + 2 = 17.

Challenge: Range (Spread of Data)

HANDLING DATA

🎯 QUICK EXAM CHALLENGE — DfE L1.HD31

Weekly sales figures are: £180, £240, £150, £310, £220. Calculate the range.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Highest = £310. Lowest = £150. Range = £310 - £150 = £160.

Challenge: Pictograms (Using a Key)

HANDLING DATA

🎯 QUICK EXAM CHALLENGE — DfE L1.HD34

On a pictogram, a circle represents 8 cars. How many circles should you draw to represent 28 cars?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

28 ÷ 8 = 3.5. Draw 3 whole circles and 1 half circle.

Challenge: Bar Charts & Scales

HANDLING DATA

🎯 QUICK EXAM CHALLENGE — DfE L1.HD33

State two things every bar chart must have to receive full marks in an exam.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

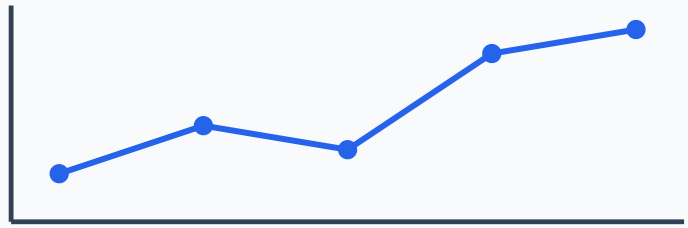
1. Equal bar widths with equal gaps. 2. Labelled axes with an evenly-spaced linear number scale starting at zero.

36. Line Graphs (Trends over Time)

HANDLING DATA

Plot Points & Join with Straight Lines

Line graphs show how continuous data changes over time (temperature, sales, patient heart rate). Time is plotted along the horizontal axis. Read points by looking across to vertical axis.



Months (Jan to May) → Overall Upward Trend

★ Golden Rule: Time always goes along the bottom horizontal axis!

⚠️ Watch Out: Join the plotted data points with neat straight ruler lines from point to point.

37. Probability Words & Scale

HANDLING DATA

Impossible to Certain

0 = Impossible (can never happen). 0.5 (1/2 or 50%) = Even Chance (e.g. flipping heads). 1 = Certain (guaranteed to happen). Unlikely is between 0 and 0.5; Likely is between 0.5 and 1.



★ Golden Rule: Probabilities on a number scale run strictly from 0 to 1!

⚠️ Watch Out: Don't use words like 'Good chance' or 'Fifty-fifty' on an exam — use 'Likely' or 'Even chance'.

38. Comparing Two Data Sets (L1)

HANDLING DATA

Compare One Average & One Range

To compare two groups for 2 marks: 1. State which group has a higher/lower mean or median. 2. State which group is more consistent by comparing their range (smaller range = more consistent).

LEVEL 1 EXAM COMPARISON TEMPLATE

1. "Team A scored higher on average (Mean 18 vs 14)."

2. "Team B was more consistent (smaller range 4 vs 10)."

★ Golden Rule: Always state the numbers AND the real-world context (sales, time)!

⚠️ Watch Out: A larger range means LESS consistent results — not better performance!

Challenge: Probability Words & Scale

HANDLING DATA

🎯 QUICK EXAM CHALLENGE — DfE L1.HD36

Match the word to the event: (a) Tomorrow will be Wednesday if today is Tuesday. (b) Rolling an 8 on a standard 6-sided die.

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

(a) Certain. (b) Impossible.

Challenge: Line Graphs (Trends over Time)

HANDLING DATA

🎯 QUICK EXAM CHALLENGE — DfE L1.HD35

Between 9 am (14°C) and 1 pm (22°C), what was the rise in temperature?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Rise = 22°C - 14°C = 8°C.

Challenge: Comparing Two Data Sets (L1)

HANDLING DATA

🎯 QUICK EXAM CHALLENGE — DfE L1.HD37

Group 1: Mean = 24 kg, Range = 6 kg. Group 2: Mean = 20 kg, Range = 2 kg. Which group was more consistent in weight?

✏️ Learner Scratchpad & Calculations — Formula Working Area

✓ STEP-BY-STEP OFFICIAL MODEL ANSWER — FULL MARKS

Group 2 was more consistent because its range was smaller (2 kg compared to 6 kg).